

Prodesk IT | Digital Agency

Project Overview

Prodesk IT is a responsive digital agency landing page built using HTML, CSS, and JavaScript. The project focuses on modern frontend development concepts such as responsive layouts, Flexbox, dark mode functionality, reusable CSS variables, hover animations, and semantic HTML structure.

Technologies Used

- HTML5
 - CSS3
 - JavaScript (Vanilla JS)
 - Font Awesome Icons
 - Google Fonts
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HTML Concepts Used

1. HTML Boilerplate Structure

```
<!DOCTYPE html>
<html lang="en">
```

Explanation

- `<!DOCTYPE html>` → Declares HTML5 document type.
 - `<html lang="en">` → Root element of webpage and language set to English.
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2. Meta Tags

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Explanation

- `charset="UTF-8"` → Supports all characters and symbols.
- `viewport` → Makes website responsive on mobile devices.

3. External CSS Linking

```
<link rel="stylesheet" href="style.css">
```

Explanation

Connects external CSS file to HTML document.

4. External Libraries

Font Awesome

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css">
```

Explanation

Used for icons like:

- Social media icons
- Globe icon
- SEO icon

Google Fonts

```
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600&display=swap" rel="stylesheet">
```

Explanation

Imports the “Inter” font family for modern typography.

Semantic HTML Elements

Header

```
<header class="header">
```

Explanation

Represents the top section of webpage containing navigation.

Navigation Bar

```
<nav class="nav">
```

Explanation

Defines navigation links of the website.

Sections

```
<section class="hero" id="home">
```

Explanation

Used to divide webpage into logical sections:

- Hero
 - About
 - Services
 - Contact
-

Footer

```
<footer class="footer">
```

Explanation

Represents bottom section containing:

- Links
 - Social icons
 - Copyright
-

Hero Section Concepts

Video Background

```
<video autoplay muted loop class="hero__video">
```

Explanation

- `autoplay` → Starts video automatically
 - `muted` → Video has no sound
 - `loop` → Video repeats continuously
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Buttons

```
<a href="#" class="btn btn--primary">Get Started</a>
```

Explanation

Anchor tag styled as a button using CSS classes.

CSS Concepts Used

1. CSS Variables

```
:root {  
  --color-primary: #00bcd4;  
}
```

Explanation

Reusable custom variables for colors and spacing.

Benefits:

- Easy theme management
- Cleaner code
- Reusability

Usage:

```
color: var(--color-primary);
```

2. Universal Selector Reset

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

Explanation

Removes default browser spacing and ensures consistent layout sizing.

3. Smooth Scrolling

```
html {  
  scroll-behavior: smooth;  
}
```

Explanation

Creates smooth scrolling effect when clicking navigation links.

4. Flexbox Layout

```
display: flex;  
justify-content: space-between;  
align-items: center;
```

Explanation

Used for responsive layouts.

- `display: flex` → Arranges items in row/column
- `justify-content` → Horizontal alignment
- `align-items` → Vertical alignment

Used in:

- Navbar
 - Footer
 - Service cards
-

5. Fixed Navigation Bar

```
position: fixed;  
top: 0;
```

Explanation

Navbar remains visible during scrolling.

6. Backdrop Blur Effect

```
backdrop-filter: blur(10px);
```

Explanation

Creates glassmorphism blur effect behind navbar.

7. Responsive Design

```
@media (max-width: 768px)
```

Explanation

Applies styles for smaller screens like mobile devices.

8. Hover Effects

```
transform: translateY(-12px);
```

Explanation

Moves element upward when hovered.

Purpose:

- Interactive UI
 - Modern animations
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9. CSS Transitions

```
transition: all 0.3s ease;
```

Explanation

Creates smooth animations.

10. Box Shadow

```
box-shadow: 0 8px 20px rgba(0,0,0,0.06);
```

Explanation

Adds depth and card elevation effect.

11. Border Radius

```
border-radius: 10px;
```

Explanation

Creates rounded corners.

12. Object Fit

```
object-fit: cover;
```

Explanation

Makes background video fill screen proportionally.

13. Dark Mode Variables

```
body.dark {  
  --color-white: #0f172a;  
}
```

Explanation

Changes website theme colors when dark mode is enabled.

14. Flex Wrap

```
flex-wrap: wrap;
```

Explanation

Moves items to next line on smaller screens.

JavaScript Concepts Used

1. DOM Selection

```
const toggle = document.getElementById("theme-toggle");
```

Explanation

Selects toggle switch element from HTML.

2. Event Listener

```
toggle.addEventListener("change", () => {
```

Explanation

Runs function whenever toggle switch changes.

3. Class Toggling

```
document.body.classList.toggle("dark");
```

Explanation

Adds/removes `dark` class dynamically.

Used for:

- Dark mode activation
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4. Conditional Statements

```
if (document.body.classList.contains("dark"))
```

Explanation

Checks whether dark mode class exists.

UI/UX Features Implemented

- Responsive Navigation
 - Dark Mode Toggle
 - Video Hero Section
 - Smooth Scrolling
 - Interactive Hover Effects
 - Responsive Service Cards
 - Social Media Footer
 - Modern Typography
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Project Architecture

The project follows Separation of Concerns:

Technology	Purpose
HTML	Structure
CSS	Styling
JavaScript	Functionality

Frontend Concepts Learned

- Semantic HTML
- Responsive Web Design

- Flexbox Layout
 - CSS Variables
 - Media Queries
 - DOM Manipulation
 - Event Handling
 - Dark Mode
 - Hover Animations
 - Modern UI Design
 - Website Structuring
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